

Time series decomposition

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Before you start

This is an in-class lab: work through the material below at your own pace (in pairs or small groups is fine) — I'll walk around to help, and we'll stop along the way to discuss as a group.

This builds on the moving averages, classical decomposition, X11/SEATS and STL material from the [decomposition chapter](#). Today you apply it: decompose real series into trend-cycle, seasonal and remainder components, compare methods, and interpret what you get.

```
library(fpp3)
library(tidyverse)
```

Cheat sheet: decomposition methods

Method	Syntax
Classical	<code>model(classical_decomposition(variable, type = "additive"))</code>
X11	<code>model(X_13ARIMA_SEATS(variable ~ x11()))</code>
SEATS	<code>model(X_13ARIMA_SEATS(variable ~ seats()))</code>
STL	<code>model(STL(variable ~ trend(window = ...) + season(window = ...), robust = ...))</code>

After fitting, use `components()` to extract the trend-cycle/seasonal/remainder columns, then `autoplot()` to plot them.

Task 1: Offshore wind power

Use the `windpower` data from `ban430data` (the same one from the [Time series graphics lab](#)) again — this time for decomposition.

```
# install.packages("remotes")
remotes::install_github("holleland/ban430data")
library(ban430data)
data(windpower)
```

- Create a tsibble containing the **daily** wind power data from **Sørlig Nordsjø 2** only (aggregate `windpower` from hourly to daily with the mean, same as in the graphics lab).
- Decompose the series into trend-cycle T_t , season S_t and remainder R_t , using a suitable decomposition method. Which method(s) from the cheat sheet above are *not* usable here, and why? Why did you choose the method you did?
- Statistics agencies and researchers often quantify *how* seasonal or trending a series is with **STL features**, rather than just eyeballing a decomposition (a preview of Chapter 4 - see [Section 4.3 of the textbook](#) for the details). Aggregate `windpower` to **monthly** data for **both** locations (like `wp_month` in the graphics lab), then run:

```
windpower %>%
  group_by_key() %>%
  index_by(yearmonth = yearmonth(datetime)) %>%
  summarise(powerprod = mean(powerprod, na.rm = TRUE)) %>%
  features(powerprod, feat_stl)
```

Which location has the stronger `seasonal_strength_year`? Does that match what `gg_season()/gg_subseries()` suggested in the graphics lab? Does `trend_strength` agree with what you found in (b)?

Task 2: Norwegian wholesale and retail sales index

The data is a monthly index for *Retail trade, except of motor vehicles and motorcycles*, from Statistics Norway (Jan 2000 onwards). It's the `wholesale_raw_lab` object from `ban430data` (same package as `windpower` above).

```
library(ban430data)
data(decomposition)

wholesale <- wholesale_raw_lab
head(wholesale, 3)
```

```
# A tibble: 3 x 2
  month   `wholesale and retail sales index`
  <chr>                                <dbl>
1 2000M01                                39.9
2 2000M02                                38.9
3 2000M03                                42.4
```

- a) Convert the `month` column to a `yearmonth` type and turn `wholesale` into a tsibble.
- b) Make a time plot.
- c) Decompose the series using the classical, X11, SEATS and STL methods. Can you detect any prominent differences between the methods?
- d) Try adjusting the trend and season windows of the STL (default values are 21 and 11 respectively). What happens?
- e) Using your method of choice, plot the seasonally-adjusted time series.
- f) Statistics Norway publishes its own seasonally-adjusted version of this exact series - it's the `wholesale_sa` object in `ban430data` (same table, same industry filter, official `VerdiinSesong` content code). Plot it alongside your own seasonally-adjusted series from (e). Do they agree?
- g) Using your method of choice, plot the detrended series.
- h) *Optional*: Implement your own additive classical decomposition by hand (moving averages + seasonal averaging), and compare to `classical_decomposition()`. (See [Chapter 3 exercises](#), [exercise 6](#) if you want to check your approach against a worked solution.)

Wrap-up discussion

For both series today: did the different decomposition methods roughly agree, or did they tell noticeably different stories? When might you prefer STL over classical decomposition, or vice versa? What made X11/SEATS unsuitable for the daily wind power data?